MODULE 2

BASIS, DIMENSION and RANK of a MATRIX

LESSON 1: RANK OF A MATRIX

Linearly Independence

To decide if a set of m-vectors $\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$ are linearly independent, we have to solve the vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$$

We know that $\mathbf{x} = \mathbf{0}$ is one solution, the *trivial solution*. Are there other (non-trivial) solutions?

If yes, then the vectors are *linearly dependent*. We can use a non-trivial solution to express one vector as a linear combination of the others.

If no, then the vectors are *linearly independent*

A linear system of m equations is the same as a single vector equation of m-vectors. We may therefore re-write a vector equation as a linear system, and also re-write a linear system as a vector equation.

Example

Write the following linear system as a vector equation:

$$\begin{array}{rcrcrcl} 2x_1 & + & 2x_2 & - & x_3 & = & 0 \\ 4x_1 & & & + & 2x_3 & = & 0 \\ & & 6x_2 & - & 3x_3 & = & 0 \end{array}$$

Solution

We re-write the three equations as one equation of 3-vectors:

$$x_1\begin{pmatrix}2\\4\\0\end{pmatrix} + x_2\begin{pmatrix}2\\0\\6\end{pmatrix} + x_3\begin{pmatrix}-1\\2\\-3\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

We may write this as

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3 = \mathbf{0}$$

Note that $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are the columns of the coefficient matrix of the linear system, and $\mathbf{0}$ is the last (augmented) column of the augmented matrix.

CRITERION FOR LINEARLY INDEPENDECE

Theorem

Let $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ be n-vectors, and let A be the $n \times n$ matrix with these vectors as columns. Then $\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$ are linearly independent if and only if

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{vmatrix} \neq 0$$

Idea for proof: The linear system $A\mathbf{x} = \mathbf{0}$ has a unique solution (that is, only the trivial solution) if and only if $\det(A) \neq 0$.

Example

Show that $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are linearly independent when

$$\mathbf{a}_1 = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} 2 \\ 0 \\ 6 \end{pmatrix}, \quad \mathbf{a}_3 = \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$$

Solution

Since we have

$$\begin{vmatrix} 2 & 2 & -1 \\ 4 & 0 & 2 \\ 0 & 6 & -3 \end{vmatrix} = (-4) \cdot 0 + (-2) \cdot 12 = -24 \neq 0$$

it follows that the vectors are linearly independent.

RANK OF A MATRIX

Let A be any m×n matrix. Then A consists of n column vectors $a_1, a_2, \ldots, a_n$, which are m-vectors.

Definition The rank of A is the maximal number of linearly independent column vectors in A, i.e. the maximal number of linearly independent vectors among $\{ a_1, a_2, \ldots, a_n \}$. If A = 0, then the rank of A is 0.

Note: We write rk(A) for the rank of A. Note that we may compute the rank of any matrix — square or not.

Let us first see how to compute the rank of a 2×2 matrix:

Example

The rank of a 2×2 matrix $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is given by:

- $\mathrm{rk}(A) = 2$ *if* $\det(A) = ad - bc \neq 0$, *since both column vectors are independent in this case*
- $\mathrm{rk}(A) = 1$ *if* $\det(A) = 0$ *but* $A \neq 0 = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, *since both column vectors are not linearly independent, but there is a single column vector that is linearly independent (i.e. non-zero)*
- $\mathrm{rk}(A) = 0$ *if* $A = 0$

How to compute the rank of a matrix?

Gauss elimination

Use elementary row operations to reduce A to echelon form. The rank of A is the number of pivots or leading coefficients in the echelon form. In fact, the pivot columns (i.e. the columns with pivots in them) are linearly independent.

Note that it is not necessary to find the reduced echelon form — any echelon form will do since only the pivots matter.

Possible ranks

Counting possible number of pivots, we see that

- $\mathrm{rk}(A) \leq m$ and $\mathrm{rk}(A) \leq n$

for any $m \times n$ matrix A.

Example

Find the rank of the matrix

$$A = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 2 & 2 & 1 \end{pmatrix}$$

Solution

We use elementary row operations:

$$A = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 2 & 2 & 1 \end{pmatrix} \dashrightarrow \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & -2 & -1 \end{pmatrix}$$

Since the echelon form has pivots in the first three columns, A has rank $\mathrm{rk}(A) = 3$. *The first three columns of A are linearly independent.*

LESSON 2: BASIS and DIMENSION

Basis

Definition. Let V be a vector space. A linearly independent spanning set for V is called a basis.

Equivalently, a subset $S \subset V$ is a basis for V if any vector $\mathbf{v} \in V$ is *uniquely represented* as a linear combination

$$\mathbf{v} = r_1\mathbf{v}_1 + r_2\mathbf{v}_2 + \cdots + r_k\mathbf{v}_k,$$

where $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are distinct vectors from S and $r_1, \ldots, r_k \in \mathrm{R}$

Example:

1. Basic Standard Basis of R^n

$$\mathbf{e}_1 = (1,0,0,\ldots,0,0),\ \mathbf{e}_2 = (0,1,0,\ldots,0,0),\ldots,\ \mathbf{e}_n = (0,0,0,\ldots,0,1)$$

2. Matrix of 2x2 $M_2 = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} \begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix} \begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix} \begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}$

3. A polynomial $1 + x + x^2 + x^3 + \cdots + x^{n-1}$ form a basis for

$$P_n = a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-2} \text{ where } a_i \in R$$

The infinite set $\{1, x, x^2, x^3, \ldots, x^{n-1}\}$ is a basis for P, the space of all polynomials.

Theorem 1:

Every basis for the vector space R^n consists of n vectors.

Theorem 2:

For any vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n \in \mathrm{R}$ the following conditions are equivalent:

(i) $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for R^n ;

(ii) $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a spanning set for R^n

(iii) $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a linearly independent set

Theorem 3:

Any vector space V has a basis. All bases for V are of the same cardinality.

Definition. The dimension of a vector space V, denoted dim V, is the cardinality of its bases.

Remark. By definition, two sets are of the same cardinality if there exists a one-to-one correspondence between their elements.

For a finite set, the cardinality is the number of its elements.

For an infinite set, the cardinality is a more sophisticated notion. For example, Z and R are infinite sets of different cardinalities while Z and Q are infinite sets of the same cardinality.

Examples. • $\dim \mathbb{R}^n = n$

• $\mathcal{M}_{2,2}(\mathbb{R})$: the space of 2×2 matrices
$\dim \mathcal{M}_{2,2}(\mathbb{R}) = 4$

• $\mathcal{M}_{m,n}(\mathbb{R})$: the space of $m\times n$ matrices
$\dim \mathcal{M}_{m,n}(\mathbb{R}) = mn$

• $\mathcal{P}_n$: polynomials of degree less than n
$\dim \mathcal{P}_n = n$

• $\mathcal{P}$: the space of all polynomials
$\dim \mathcal{P} = \infty$

• $\{\mathbf{0}\}$: the trivial vector space
$\dim \{\mathbf{0}\} = 0$

Example 1; Find the dimension of the plane x + 2z = 0 in the R^3.

The general solution of x +2z = 0 is ….
x = -2z we let z = s and t =y thus we have

$$\begin{cases} x = -2s \\ y = t \\ z = s \end{cases} \quad where\ s,t \in R$$

Now, (x, y, z) = (-2s , t, s) = t (0,1,0) + s (-2,0,1) . Hence the plane span for the vectors v_1 = (0,1,0) and v_2 = (-2,0,1)
These vectors are linearly independent as they are not parallel with each other. Thus $\{v_1, v_2\}$ is a basis for R^3 thus the dimension of the plane is 2.

How to find a basis ?

Theorem 4

Let *S* be a subset of a vector space *V*. Then the following conditions are equivalent:
(i) *S* is a linearly independent spanning set for *V*, i.e., a basis;
(ii) *S* is a minimal spanning set for *V*;
(iii) *S* is a maximal linearly independent subset of *V*.
"Minimal spanning set" means "remove any element from this set, and it is no longer a spanning set".
"Maximal linearly independent subset" means "add any element of *V* to this set, and it will become linearly dependent".

Theorem 5

Let *V* be a vector space. Then
(i) any spanning set for *V* can be reduced to a minimal spanning set;
(ii) any linearly independent subset of *V* can be extended to a maximal linearly independent set.

Equivalently, any spanning set contains a basis, while any linearly independent set is contained in a basis.

Corollary 6

A vector space is finite-dimensional if and only if it is spanned by a finite set.

Approach 1. Get a spanning set for the vector space, then reduce this set to a basis.

Proposition 7

Let $v_0, v_1, \ldots, v_k$ be a spanning set for a vector space V. If v_0 is a linear combination of vectors $v_1, \ldots, v_k$ then $v_1, \ldots, v_k$ is also a spanning set for V.

Indeed, if $v_0 = r_1 v_1 + \cdots + r_k v_k$, then

$$= t_0 v_0 + t_1 v_1 + \cdots + t_k v_k$$

$$= (t_0 r_1 + t_1) v_1 + \cdots + (t_0 r_k + t_k) v_k.$$

Approach 2. Build a maximal linearly independent set adding one vector at a time.

If the vector space V is trivial, it has the empty basis. If $V \neq \{0\}$, pick any vector $v_1 \neq 0$.

If v_1 spans V, it is a basis. Otherwise pick any vector $v_2 \in V$ that is not in the span of v_1.

If v_1 and v_2 span V, they constitute a basis. Otherwise pick any vector $v_3 \in V$ that is not in the span of v_1 and v_2.

And so on…

Problem:
Find a basis for the vector space V
spanned by vectors $w_1 = (1, 1, 0)$, $w_2 = (0, 1, 1)$, $w_3 = (2, 3, 1)$, and $w_4 = (1, 1, 1)$.

Answer:
To pare this spanning set, we need to find a relation of the form $r_1 w_1 + r_2 w_2 + r_3 w_3 + r_4 w_4 = 0$, where $r_i \in R$ are not all equal to zero. Equivalently,

$$\begin{pmatrix} 1 & 0 & 2 & 1 \\ 1 & 1 & 3 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

To solve this system of linear equations for r_1, r_2, r_3, r_4, we apply row reduction.

$$\begin{pmatrix} 1 & 0 & 2 & 1 \\ 1 & 1 & 3 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\to \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{(reduced row echelon form)}$$

$$\begin{cases} r_1 + 2r_3 = 0 \\ r_2 + r_3 = 0 \\ r_4 = 0 \end{cases} \iff \begin{cases} r_1 = -2r_3 \\ r_2 = -r_3 \\ r_4 = 0 \end{cases}$$

General solution: $(r_1, r_2, r_3, r_4) = (-2t, -t, t, 0),\ t \in \mathbb{R}$.
Particular solution: $(r_1, r_2, r_3, r_4) = (2, 1, -1, 0)$.

Problem 2

Find a basis for the vector space V
spanned by vectors $w_1 = (1, 1, 0)$, $w_2 = (0, 1, 1)$, $w_3 = (2, 3, 1)$, and $w_4 = (1, 1, 1)$.

Answer:

We have obtained that $2w_1 + w_2 - w_3 = 0$. Hence any of vectors w_1, w_2, w_3 can be dropped. For instance, $V = \text{Span}(w_1, w_2, w_4)$.

Let us check whether vectors w_1, w_2, w_4 are linearly independent

$$\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1 \neq 0.$$

They are!!! It follows that $V = \mathbb{R}^3$ and
$\{w_1, w_2, w_4\}$ is a basis for V.

Problem 3

Vectors $v_1 = (0, 1, 0)$ and $v_2 = (-2, 0, 1)$ are linearly independent.

Extend the set $\{v_1, v_2\}$ to a basis for R .

Our task is to find a vector v_3 that is not a linear combination of v_1 and v_2.

Then $\{v_1, v_2, v_3\}$ will be a basis for R^3.
Hint 1. v_1 and v_2 span the plane $x + 2z = 0$. The vector $v_3 = (1, 1, 1)$ does not lie in the plane $x + 2z = 0$, hence it is not a linear combination of v_1 and v_2. Thus $\{v_1, v_2, v_3\}$ is a basis for R^3

Hint 2. At least one of vectors $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, and $e_3 = (0, 0, 1)$ is a desired one.

Let us check that $\{v_1, v_2, e_1\}$ and $\{v_1, v_2, e_3\}$ are basis for R^3.

$$\begin{vmatrix} 0 & -2 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 1 \neq 0, \qquad \begin{vmatrix} 0 & -2 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2 \neq 0.$$

end

ACTIVITY #2 : RANK OF A MATRIX, BASIS and DIMENSION

Direction: Write your answer in a an A4/ Shor/ Long Bond paper with your PRINTED NAME And Signature at the bottom right. (Be sure to answer it neatly. If you can write your answer in Microsoft Word much better save a file in this format SURNAME-2BASIS. If your answer is in written take a photo and save a file name SURNAMAE-2BASIS. Submit your output in the drop box created in EDMODO.

A. Determine the rank of a matrix. Show your solution

1. $\begin{pmatrix} 1 & 3 \\ -2 & 2 \end{pmatrix}$

2. $\begin{pmatrix} 2 & 6 & -2 \\ 3 & -2 & 8 \end{pmatrix}$

3. $\begin{pmatrix} 2 & -2 & 4 \\ 4 & 1 & -2 \\ 6 & -1 & 2 \end{pmatrix}$

4. $\begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}$

5. $(-2 \quad 3 \quad 1)$

B. Basis and Dimension

For 6-8 Let $a_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, a_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $a_3 = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$

6. Express a_3 **as a linear combination of** a_1 and a_2 **if possible.**

7. Is $\{a_1, a_2, a_3\}$ **is a basis of** R^2 **? why or why not?**

8. Is $\{a_2, a_3\}$ **is a basis of** R^2 **? why or why not ?**

For 9-10 Let $u_1 = (4, 2.5), u_2 = (3, -1, -2)$ and $u_3 = (6, 2, 0)$

9 . Is $\{u_1, u_2, u_3\}$ **is a basis of** R^3 **? Justify your answer.**

10. Is it possible that the vector (9,5,2) as a linear combination of $u_1, u_2,$ and u_3**. Justify your answer without showing your solution.**